LOCAL ADVECTION OF MOMENTUM, HEAT, AND MOISTURE IN MICROMETEOROLOGY

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(Received 6 May, 1974)

Abstract. The local advection of momentum, heat and moisture in micrometeorology due to a horizontal inhomogeneity in surface conditions is numerically investigated by a higher-order turbulence closure model which includes equations for the mean quantities, turbulent fluxes, and the viscous dissipation rate. The application of the two-dimensional model in this paper deals with the simulation of the flow from an extensive smooth dry area to a grassy wet terrain. The mean wind speed, temperature, and humidity distributions in the resulting internal boundary layer downstream of the surface discontinuity are determined such that the energy and moisture balances at the Earth's surface are satisfied.

Numerical calculations of the mean temperature and humidity profiles are compared with available observed ones. The results include the advective effects on turbulent flux distributions, surface energy balance, evaporation rate, and Bowen ratio. The sensitivity of the predicted mean profiles and turbulent flux distributions to the surface relative humidity, thermal stratification, and the roughness change is discussed.

1. Introduction

In micrometeorology, the well-known flux-profile relationships (Businger *et al.*, 1971) are based on the assumptions that the surface layer can be treated as horizontally homogeneous and steady in the mean. This permits a one-dimensional treatment of the problem. Real atmospheric flows, where horizontal inhomogeneity is a common occurrence, do not generally correspond to this ideal situation. For example, in land-lake interface problems, and in flows over finite areas of irrigated land, abrupt changes in surface conditions such as in roughness, temperature, and humidity occur simultaneously, and the profiles are no longer in equilibrium. In micrometeorology, the horizontal transport of momentum, heat, and moisture resulting from surface inhomogeneities is known as local advection. Since the flow structure depends on the distance downstream of the discontinuity as well as the height above the surface, these problems are treated as two-dimensional.

In addition to providing quantitative estimates of advective errors in micrometeorological field experiments (see, e.g., Dyer and Pruitt, 1962; Blad and Rosenberg, 1974), the theory of local advection has important applications in agronomy, biology, hydrology, and air pollution. For instance, when a finite extent of dry land is heavily irrigated, the energy balance at the Earth's surface is altered such that a larger proportion of the net radiation is spent in evaporation, and a smaller proportion on heat transfer to the atmosphere and soil. The humidity near the surface is thereby increased

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Boundary-Layer Meteorology 7 (1974) 331–348. *All Rights Reserved*
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while the temperature is decreased. This may lead to the formation of an advective inversion near the surface. Also, the reduction in surface temperature due to irrigation in turn reduces the emission of long-wave radiation from the surface. Information on the modification of the surface energy balance and evaporation rates over irrigated lands and reservoirs, and the change in microclimate due to local advection has vital applications in agriculture and irrigation engineering.

In this paper, a higher-order turbulence closure model is outlined to study the local advection of momentum, heat, and moisture in the lower atmosphere. The model consists of exact equations for the mean quantities, and a full set of turbulence equations which are closed approximately. The approach is generally similar to the mean Reynolds stress closure models proposed by Donaldson (1973) and Mellor (1973), among others. However, the present model does not require *a priori* information on the behavior of a length scale since the latter is determined by the system itself. A discussion of the model and the closure approximations is given by Wyngaard *et al.* (1974) and Wyngaard and Coté (1974).

The present application of the two-dimensional model deals with the flow normal to a surface discontinuity at $x=0$ from a smooth dry area to a grassy wet terrain. The essential problem is to determine the mean wind speed, temperature, and humidity fields in the internal boundary layer such that the energy and moisture balances at the Earth's surface are satisfied.

Several theoretical and experimental studies of this problem are available in the literature. These include Timofeev (1954), Philip (1959), de Vries (1959), Rider *et al.* (1963), Dyer and Crawford (1965), and Morgan *et al.* (1970). The theoretical treatments, based on assumed power forms for the wind speed and the eddy diffusivity distributions, generally neglect the effects of roughness change and thermal stratification on the air flow, and assume that the wet surface is saturated. Taylor (1971) emphasized the importance of these effects in a numerical model based on a mixing-length approach with empirical forms for nondimensional wind shear, temperature and humidity gradients. In his model, the variations of surface fluxes of sensible and latent heat were specified *a priori* for $x>0$, instead of being determined from the surface energy balance.

In this study, numerical calculations of the mean temperature and humidity profiles (covering a distance of 20 m from the discontinuity) are compared with the available experimental data of Rider *et al.* (1963). The results include the advective effects on turbulent flux distributions, surface energy balance, evaporation rate, and Bowen ratio. The present model accounts for the surface relative humidity, buoyancy, and the roughness changes. The sensitivity of the predicted mean profiles and turbulent flux distributions to these factors has been analyzed.

2. Model Equations

The governing equations for the basic variables utilized to investigate the local advection of momentum, heat, and moisture in the surface layer are given in this section.

Under the assumptions discussed by Rao *et al.* (1974), the equations of the steady-state two-dimensional mean field in the internal boundary layer can be written as follows:

$$\begin{aligned} U\frac{\partial U}{\partial x} + W\frac{\partial U}{\partial z} &= -\frac{\partial}{\partial z}\overline{uw}, \\ U\frac{\partial \Theta}{\partial x} + W\frac{\partial \Theta}{\partial z} &= -\frac{\partial}{\partial z}\overline{\theta w}, \\ U\frac{\partial Q}{\partial x} + W\frac{\partial Q}{\partial z} &= -\frac{\partial}{\partial z}\overline{qw}, \\ \frac{\partial U}{\partial x} + \frac{\partial W}{\partial z} &= 0. \end{aligned} \tag{1}$$

Here the mean wind vector, mean potential temperature, and the mean specific humidity are denoted by $U_i=(U, 0, W)$, Θ, and Q, respectively; the corresponding fluctuating components are indicated by $u_i=(u, v, w)$, θ, and q.

The transport equations for turbulent fluxes of momentum, heat, and moisture are, respectively,

$$\begin{aligned} U_j(\overline{u_iu_k})_{,j} + \overline{u_ju_k}\,U_{i,j} + \overline{u_ju_i}\,U_{k,j} - \frac{g}{T_0}(\overline{u_k\theta_v}\,\delta_{3i} + \overline{u_i\theta_v}\,\delta_{3k}) \\ = (\overline{u_iu_ku_j})_{,j} - (\overline{u_kp_{,i}} + \overline{u_ip_{,k}}) - 2\bar{\varepsilon}\frac{\delta_{ik}}{3} \end{aligned} \tag{2}$$

$$U_j(\overline{\theta u_i})_{,j} + \overline{\theta u_j}\,U_{i,j} + \overline{u_ju_i}\,\Theta_{,j} - \frac{g}{T_0}\overline{\theta\theta_v}\,\delta_{3i} = (\overline{\theta u_iu_j})_{,j} - \overline{p_{,i}\theta} \tag{3}$$

$$U_j(\overline{qu_i})_{,j} + \overline{qu_j}\,U_{i,j} + \overline{u_ju_i}\,Q_{,j} - \frac{g}{T_0}\overline{q\theta_v}\,\delta_{3i} = (\overline{qu_iu_j})_{,j} - \overline{p_{,i}q}, \tag{4}$$

where differentiation is indicated by a comma and repeated indices denote summation. In the above, $\bar{\varepsilon}$ is the viscous dissipation rate of turbulent kinetic energy ($\frac{1}{2}\overline{u_iu_i}$), p is the fluctuating kinematic pressure, $\theta_v=\theta+0.61\,T_0q$ is the fluctuating virtual potential temperature, T_0 is a constant representative value of the mean temperature, and g is the acceleration due to gravity.

The equations for the temperature and humidity variances are

$$U_j\overline{\theta^2}_{,j} + 2\overline{u_j\theta}\,\Theta_{,j} = -(\overline{\theta^2u_j})_{,j} - 2\bar{\varepsilon}_\theta \tag{5}$$

$$U_j\overline{q^2}_{,j} + 2\overline{u_jq}\,Q_{,j} = -(\overline{q^2u_j})_{,j} - 2\bar{\varepsilon}_q, \tag{6}$$

where $\bar{\varepsilon}_\theta$ and $\bar{\varepsilon}_q$ are the rates of dissipation of $\overline{\theta^2}$ and $\overline{q^2}$, respectively, by molecular effects. The final equation required is that for $\overline{q\theta}$:

$$U_j(\overline{q\theta})_{,j} + \overline{u_jq}\,\Theta_{,j} + \overline{u_j\theta}\,Q_{,j} = -(\overline{\theta qu_j})_{,j} - 2\bar{\varepsilon}_{q\theta}, \tag{7}$$

where $\bar{\varepsilon}_{q\theta}$ is the rate of molecular destruction of $\overline{q\theta}$.

All terms on the right-hand side of Equations (2) to (7) are unknowns. To close the equations, we have to approximate the unknowns in terms of other basic variables for which we have or can write equations. These approximations, which constitute the higher-order closure technique, are discussed in detail by Lumley and Khajeh-Nouri (1974), Wyngaard *et al.* (1974) and Wyngaard and Coté (1974). Therefore, only the final expressions are given below.

A dynamical equation for the turbulent energy dissipation rate can be written and, as shown in Wyngaard *et al.* (1974), approximated as

$$U_j\bar{\varepsilon}_{,j} + 4\frac{\bar{\varepsilon}}{\overline{u_iu_i}}(\bar{\varepsilon} - aP) = -(\overline{u_j\varepsilon})_{,j}, \tag{8}$$

where $a=0.5$ and P is the production rate of turbulent kinetic energy. In principle, the equations for the molecular destruction rates of $\overline{\theta^2}$, $\overline{q^2}$, and $\overline{q\theta}$ also can be written, but we simply used

$$\bar{\varepsilon}_\theta = b\frac{\overline{\theta^2}}{\tau}, \qquad \bar{\varepsilon}_q = b\frac{\overline{q^2}}{\tau}, \qquad \bar{\varepsilon}_{q\theta} = b\frac{\overline{q\theta}}{\tau}, \tag{9}$$

where $b=1.4$ and $\tau=\overline{u_iu_i}/\bar{\varepsilon}$ is a turbulence relaxation time determined by the model itself.

The pressure covariance terms in Equations (2) to (4) are modeled as

$$\begin{aligned} -(\overline{u_kp_{,i}} + \overline{u_ip_{,k}}) + 2(\overline{u_jp})_{,j}\frac{\delta_{ik}}{3} &= -\left(\overline{u_iu_k} - \overline{u_iu_i}\frac{\delta_{ik}}{3}\right)\frac{c_{ik}}{\tau} \\ -\overline{\theta p_{,i}} &= -d_i\frac{\overline{\theta u_i}}{\tau} \qquad \text{(no sum)} \\ -\overline{qp_{,i}} &= -d_i\frac{\overline{qu_i}}{\tau}, \end{aligned} \tag{10}$$

where the constants $c_{ii}=6.7$, c_{ik} $(i\neq k)=13.2$, $d_1=4.4$ and $d_3=9.7$. Finally, the turbulent flux divergence terms in Equations (3) to (8) are approximated by a simple gradient transport model as

$$\overline{fu_j} = -a_tf^*_{,m}\overline{u_mu_j}\,\tau, \tag{11}$$

where f^* can be $\overline{\theta u_i}$, $\overline{qu_i}$, $\overline{\theta^2}$, $\overline{q^2}$, or $\bar{\varepsilon}$. Consistent with Equation (10), the third moment $\overline{u_iu_ku_j}$ is expressed as

$$\overline{(u_iu_k + \tfrac{2}{3}p\delta_{ik})u_j} = a_t(-\overline{u_iu_k})_{,m}\overline{u_mu_j}\,\tau, \tag{12}$$

where a_t is a transport constant. We used $a_t=0.15$ except in the $\bar{\varepsilon}$-equation, where the surface-layer constraints require $a_t=(4-4a)/12.1$, as shown in Wyngaard *et al.* (1974).

Substituting the above closure approximations for the terms on the right-hand side of Equations (2) to (8), we obtain a two-dimensional model consisting of 16 coupled parabolic partial differential equations. These include four mean field equations (U, W, Θ, Q), four Reynolds stress equations ($\overline{u^2}$, $\overline{v^2}$, $\overline{w^2}$, $\overline{uw}$), two heat flux equations

($\overline{\theta w}$, $\overline{\theta u}$), two humidity flux equations ($\overline{qw}$, $\overline{qu}$), and one equation each for temperature variance ($\overline{\theta^2}$), humidity variance ($\overline{q^2}$), temperature-humidity covariance ($\overline{q\theta}$), and energy dissipation rate ($\bar{\varepsilon}$).

3. Numerical Solution, Initial and Boundary Conditions

The governing equations together with the prescribed initial and boundary conditions were numerically integrated on a CDC-6600 digital computer using a Dufort-Frankel explicit finite-difference scheme (1953), with forward-marching in the x direction. A logarithic transformation of the vertical coordinate was used to obtain a fine computational mesh near the lower boundary. The grid spacing in the x direction was uniform, with the step size determined by the considerations of stability and convergence of the numerical solution.

The lower boundary conditions were based on the equilibrium flux-profile relations defined by recent Kansas results (Businger *et al.*, 1971; Wyngaard *et al.*, 1971). These are given as follows:

At

$$
\begin{aligned}
&z = z_0, \\
&U = W = 0, \qquad \Theta = \Theta_0, \qquad Q = Q_0, \\
&\frac{\partial U}{\partial z} = \frac{u_*}{kz_0}, \qquad \frac{\partial \Theta}{\partial z} = 0.74\,\frac{T_*}{kz_0}, \qquad \frac{\partial Q}{\partial z} = 0.74\,\frac{q_*}{kz_0} \\
&\overline{uw} = -\,u_*^2, \qquad \overline{\theta w} = -\,T_* u_*, \qquad \overline{qw} = -\,q_* u_* \\
&\overline{u^2} = 4u_*^2, \qquad \overline{v^2} = \overline{w^2} = 1.75u_*^2 \\
&\overline{\theta^2} = 4T_*^2, \qquad \overline{\theta u} = 3T_* u_*, \qquad \bar{\varepsilon} = \frac{u_*^3}{kz_0} \\
&\overline{q^2} = 4q_*^2, \qquad \overline{qu} = 3q_* u_*, \qquad \overline{q\theta} = 4q_* T_*,
\end{aligned}
\tag{13}
$$

where

$$
\begin{aligned}
u_*^2 &= -\,\overline{uw} = \frac{\tau_0}{\rho} \\
T_* u_* &= -\,\overline{\theta w} = \frac{-\,H_0}{\rho c_p} \\
q_* u_* &= -\,\overline{qw} = \frac{-\,F_0}{\rho}.
\end{aligned}
$$

Here, k is the von Kármán constant (0.35), ρ is the density of air (1075 g m^{-3}), c_p is the specific heat of air at constant pressure (0.24 cal g^{-1} K^{-1}), z_0 is the surface roughness parameter, and τ_0, H_0 and F_0 are, respectively, the shear stress, heat flux, and moisture flux at the surface. Though z_0 is essentially an integration constant determined from the equilibrium mean wind profile, its value is assumed to be identical for the mean wind, temperature and humidity profiles.

Using the suffix 1 to denote the known upstream parameters, we can specify

$$z_0 = z_{01}, \qquad u_* = u_{*1}, \qquad \Theta_0 = \Theta_{01}, \qquad Q_0 = Q_{01}$$
$$T_* = T_{*1} = -\frac{H_{01}}{\rho c_p u_{*1}}, \qquad q_* = q_{*1} = -\frac{F_{01}}{\rho u_{*1}} = 0. \tag{14}$$

The upstream boundary conditions (for $x<0$) are obtained by substituting the above values in Equation (13). We assumed that the mean upstream humidity is constant with height and, therefore, the moisture flux in the upstream region is zero.

To start the computations, the initial distributions of the variables in the unstably stratified surface layer upstream of the discontinuity should be specified. The initial profiles for the present study were generated by substituting the upstream boundary conditions in the model equations and numerically integrating the latter in x (without any change in boundary conditions) until steady-state equilibrium distributions were established.

Since the characteristic time scales near the lower boundary (which are of the order of z_0/u_*) are very small compared to the external time scales, we can assume *local similarity* of the flow near the lower boundary. Under this assumption, the relations given in Equation (13) can be used to express the lower boundary conditions in terms of local parameters in the non-equilibrium flow downstream. For $x \geqslant 0$, we specify

$$z_0 = z_{02}, \qquad u_* = u_{*2}(x), \qquad \Theta_0 = \Theta_{02}(x), \qquad Q_0 = Q_{02}(x)$$
$$T_* = T_{*2}(x) = -\frac{H_{02}(x)}{\rho c_p u_{*2}(x)}, \qquad q_* = q_{*2}(x) = -\frac{F_{02}(x)}{\rho u_{*2}(x)}, \tag{15}$$

where suffix 2 is used to denote the downstream conditions.

The boundary conditions $\Theta_{02}(x)$ and $Q_{02}(x)$ are determined by the equations for energy and water balance (Decker, 1965) over the wet surface, as follows:

$$H_{02}(x) + LF_{02}(x) = R_N - G \tag{16}$$

$$Q_{02}(x) = R_0 Q_s(\Theta_{02}). \tag{17}$$

Here R_N is the net radiation, G is the soil heat flux, L is the latent heat of evaporation (585 cal g^{-1}), R_0 is the surface relative humidity (assumed to be constant) and $Q_s(\Theta_{02})$ is the saturation specific humidity at the surface temperature. The sign convention for the various energy components in Equation (16) is given by Munn (1966).

It is generally assumed that $R_N - G$ does not change significantly from the dry area to the wet area (Rider *et al.*, 1963; Dyer and Crawford, 1965). Since the latent heat flux is zero in the dry area, we can substitute $R_N - G = H_{01}$ in Equation (16). This implies that the advective effect due to the presence of the wet surface for $x \geqslant 0$ amounts to a change (with fetch) in the partition of available energy (H_{01}) between the sensible heat flux (H_{02}) and the latent heat flux (LF_{02}).

Utilizing the equilibrium flux-profile relations, the energy balance Equation (16)

can be expressed in terms of mean quantities as follows:

$$\left(\frac{\partial \Theta}{\partial z}\right)\bigg|_{z_{02}} + \frac{L}{\rho c_p}\left(\frac{\partial Q}{\partial z}\right)\bigg|_{z_{02}} = \frac{0.74 T_{*1} u_{*1}}{k z_{02} u_{*2}(x)}. \tag{18}$$

where

$$u_{*2}(x) = k z_{02}\left(\frac{\partial U}{\partial z}\right)\bigg|_{z_{02}}. \tag{19}$$

At every $x \geqslant 0$, u_{*2}, Θ_{02}, and Q_{02} can be determined by solving Equations (17), (18), and (19) together. This in turn allows determination of $T_{*2}(x)$ and $q_{*2}(x)$, and therefore all the downstream boundary conditions.

From the mean field equations and the surface energy balance equation, we can show approximately that

$$\frac{\Delta \Theta_0}{\Delta E_0} = -\frac{L}{\rho c_p}, \tag{20}$$

where $\Delta\Theta_0$ and ΔE_0 are the steps in surface temperature and absolute surface humidity between upstream and downstream at $x=0$. Knowing the upstream surface conditions, an estimate of the downstream surface conditions Θ_{02} and Q_{02} at $x=0$ can be obtained by solving Equations (17) and (20) together.

4. Results and Discussion

The results of numerical calculations covering a distance of 20 m downstream of the surface discontinuity are presented in this section. The mean temperature and humidity profiles predicted by the theory are compared with the experimental data of Rider *et al.* (1963). The latter's observation No. 6 was selected for comparison since, in this run, the upstream mean humidity was constant with height, and the data clearly showed the formation of an advective inversion.

The parameters used in the numerical study are $z_{01}=0.002$ cm, $u_{*1}=15$ cm s^{-1}, $H_{01}=8.5$ mcal cm^{-2} s^{-1}, $F_{01}=0$ g cm^{-2} s^{-1}, $E_{01}=6.34$ g m^{-3} and $z_{02}=0.14$ cm. The z_0 and E_{01} values are given by Rider *et al.* (1963), and the H_{01} and F_{01} values are consistent with their net radiation and soil heat flux data. They did not measure u_{*1}, but we estimated its value by considering the measured wind profile at $x=0$ and the surface roughness. Assuming $R_0=0.60$, the steps in surface temperature and humidity at $x=0$ are calculated from Equations (17) and (20) as $\Delta\Theta_0=21$ °C and $\Delta E_0=9.5$ g m^{-3}, respectively. These values agree well with the values calculated by Sibbons (see Rider *et al.*, 1965) for this run.

The calculated mean temperature and mean humidity profiles at different x are shown in Figures 1 and 2, where they are compared with the observations of Rider *et al.* (1963). The predicted profiles show good agreement with the observed ones. In the calculations, the relative humidity of the wet surface is assumed to be 60%. The temperature profiles clearly show the development of an advective inversion

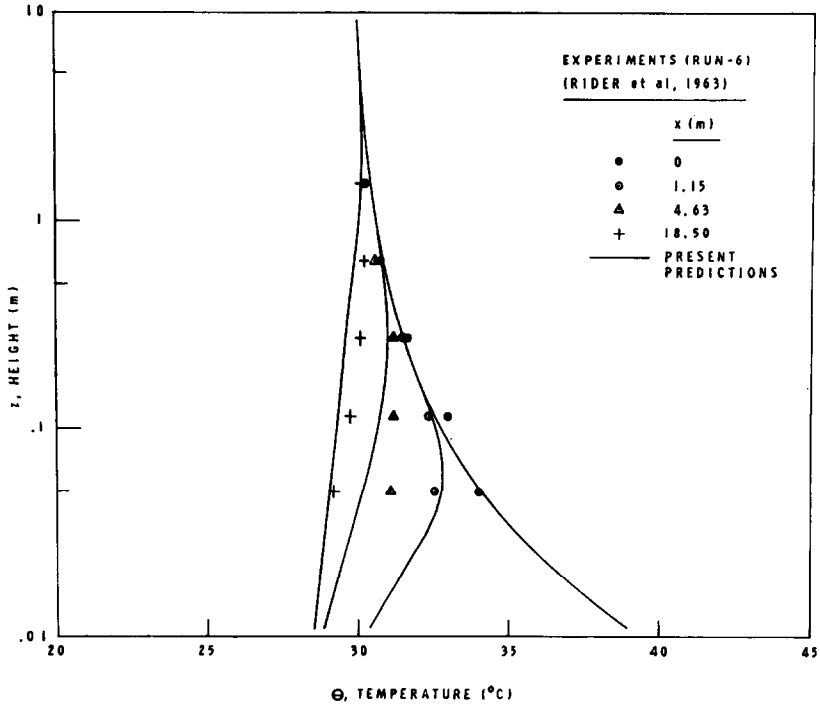


Fig. 1. Calculated mean temperature profiles – comparison with experimental data.

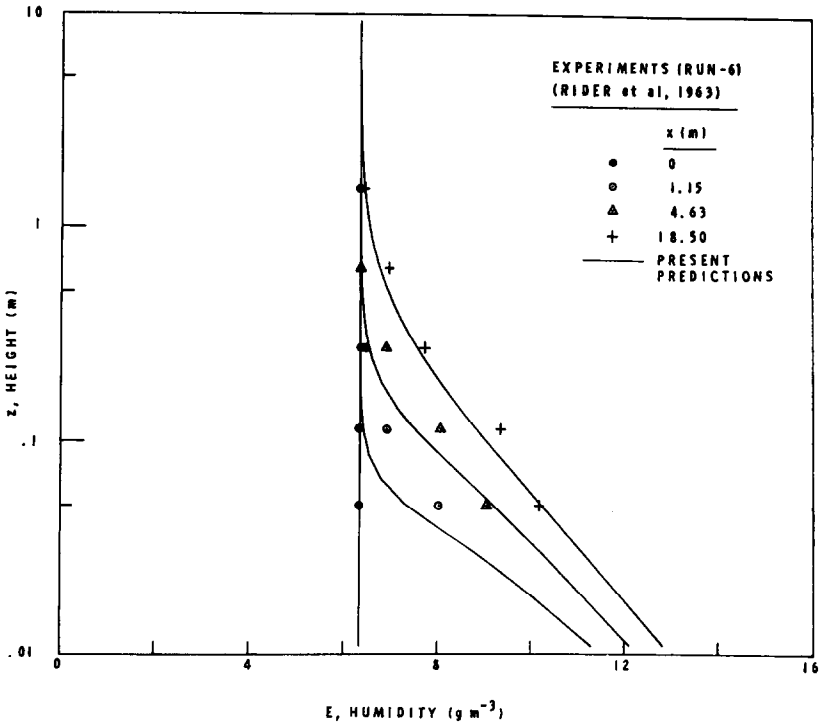


Fig. 2. Calculated mean humidity profiles – comparison with experimental data.

downstream resulting from the sharp reduction in surface temperature due to irrigation. This in turn increases the humidity of the air near the surface as shown in Figure 2.

The calculated wind profiles are given in Figure 3. The results show a deceleration of the flow near the surface as the wind moves over the aerodynamically rougher grass area downstream from a comparatively smooth upstream terrain. The detailed wind profile measurements are not available for comparison with the calculated profiles.

Figure 4 shows the advective effects on various components of the surface energy balance represented by Equation (16). Most of the variation in the sensible and latent heat fluxes at the surface occurs within a few meters from the discontinuity. The sensible heat flux at the surface abruptly changes sign from positive (unstable stratification) to negative (stable stratification) at $x=0$ due to the presence of moisture at the surface and the resulting demand for energy consumption by evaporation, which exceeds the available energy. This leads to a loss of sensible heat by the air near the surface, and the formation of an advective inversion.

The variation of the surface Bowen ratio (B_0), defined as the ratio of sensible to latent heat flux at the surface, is also shown in Figure 4. The maximum $|B_0|$ occurs near the surface discontinuity where the sensible heat flux proportionately is the largest. Far downwind, the sensible heat flux becomes negligibly small, and B_0 approaches zero. In this limit, the latent heat flux is equal to the available energy, i.e., to $(R_N - G)$, and the temperature profiles approach the neutral adiabatic distributions. The downstream variation of the surface shear stress ratio $\tau_{02}(x)/\tau_{01}$, with τ_0 defined as in Equation (13), is also given in this figure.

The effect of surface relative humidity on the calculated mean temperature and mean humidity profiles is shown in Figures 5 and 6, respectively. The figures show three sets of data corresponding to $R_0 = 60$, 80, and 100% (saturation). As R_0 increases, the air temperature decreases while the humidity increases. Based on eddy diffusivities, an advection model for this flow was given by Rider *et al.* (1963), in which the surface was assumed to be saturated because the grass was frequently irrigated, ensuring that water was freely available at the surface during the experiment. This assumption does not take into account the ability of a vegetative cover to control the water loss by internal resistance to evapotranspiration. Monteith (see Rider *et al.*, 1965) pointed out that even when the soil is at field capacity, a crop does not behave like a free water surface. Our calculations show that the assumption of saturation at the surface underpredicts temperatures and overpredicts humidities.

The calculated variations of the surface evaporation rate, $S_0(x)$, and the surface Bowen ratio, $B_0(x)$, are shown in Figure 7 for the three values of R_0. The maximum evaporation rates occur near $x=0$. In the first one meter, the average evaporation rates range from 1 mm hr^{-1} (with $R_0 = 60\%$) to 1.3 mm hr^{-1} (with surface at saturation).

In the advection model given by Rider *et al.* (1963), the effects of the roughness change and thermal stratification on the flow were not included. The neglect of these effects was suggested as a possible cause for discrepancies between their calculations

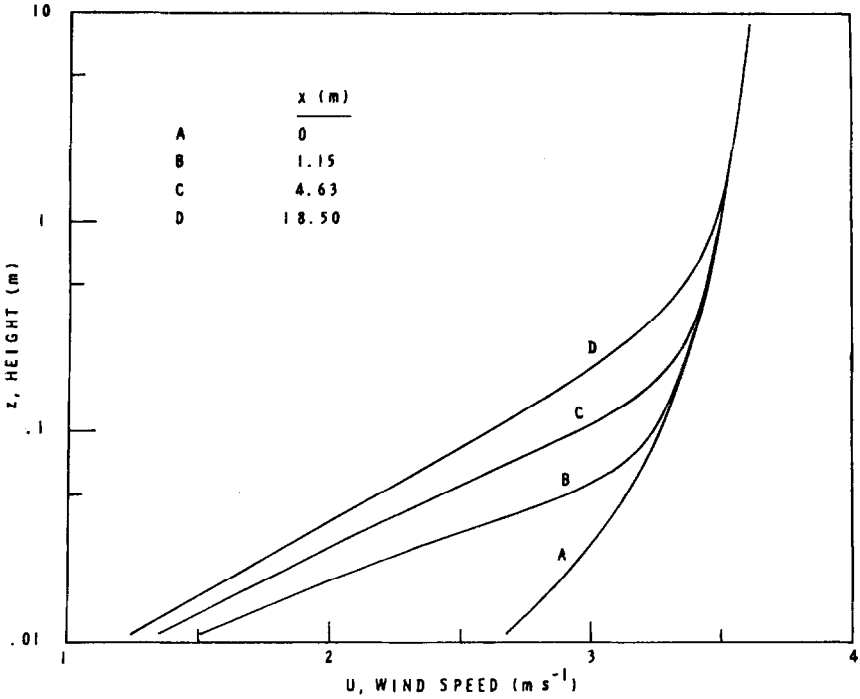


Fig. 3. Calculated mean wind profiles.

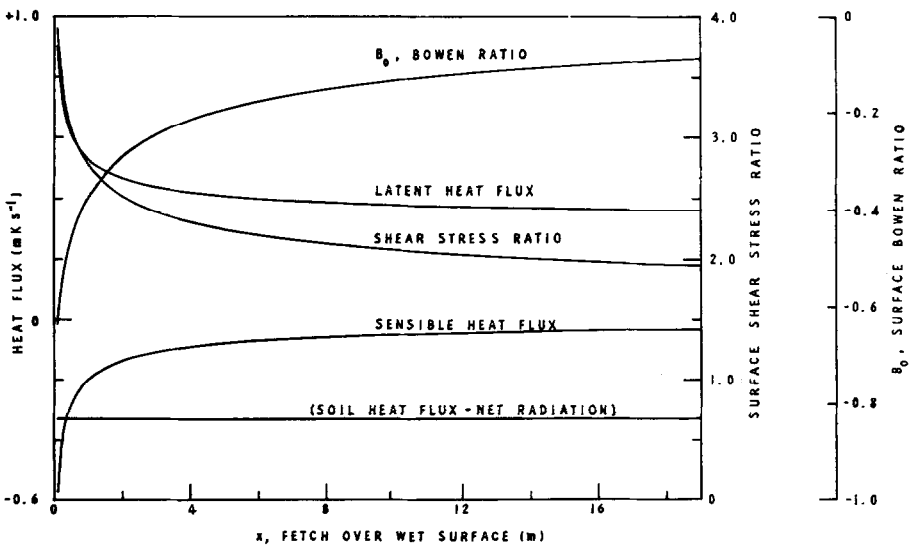


Fig. 4. Energy balance and variation of Bowen ratio and shear stress over wet surface.

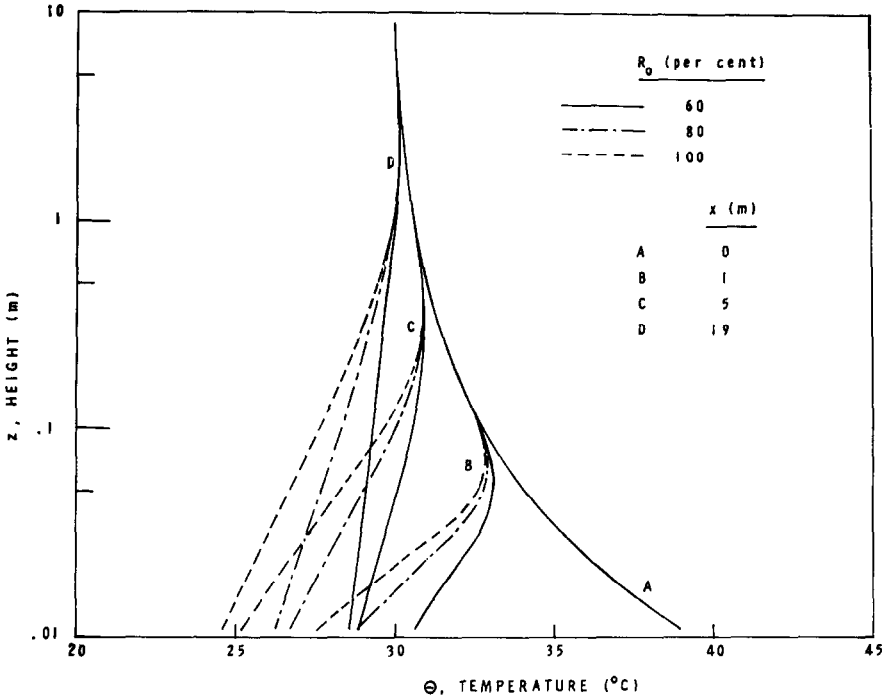


Fig. 5. Effect of variation of surface relative humidity on calculated mean temperature profiles.

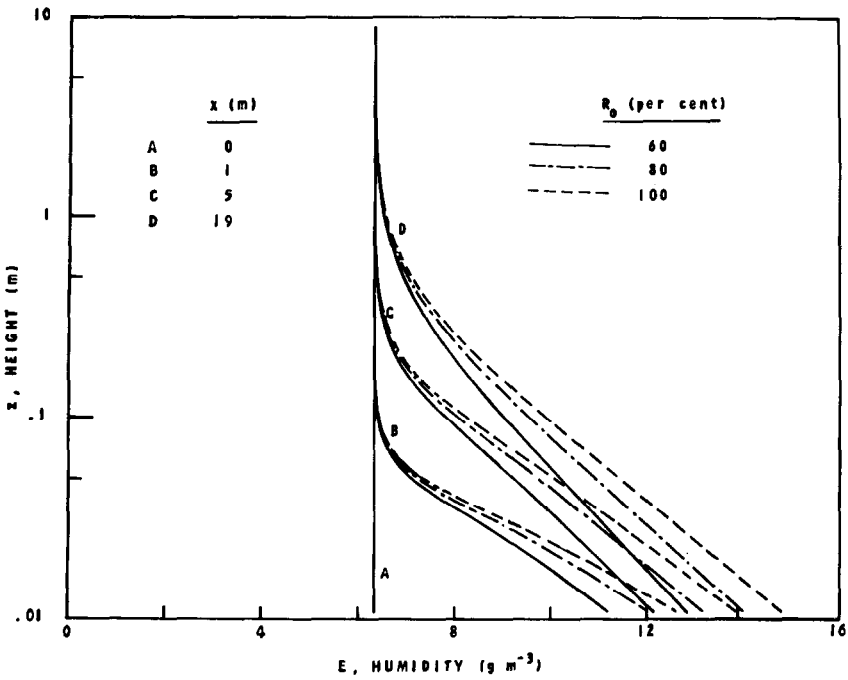


Fig. 6. Effect of variation of surface relative humidity on calculated mean humidity profiles.

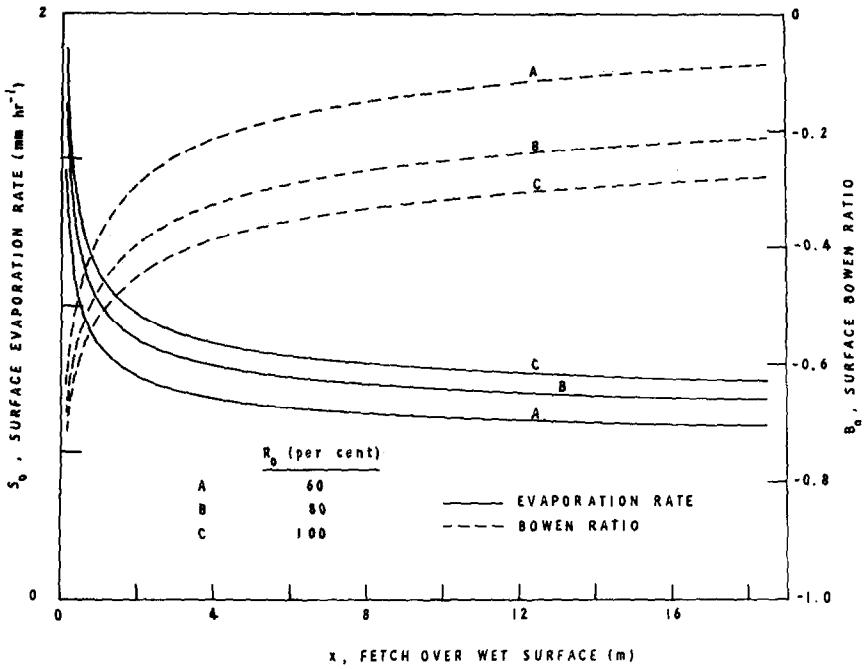
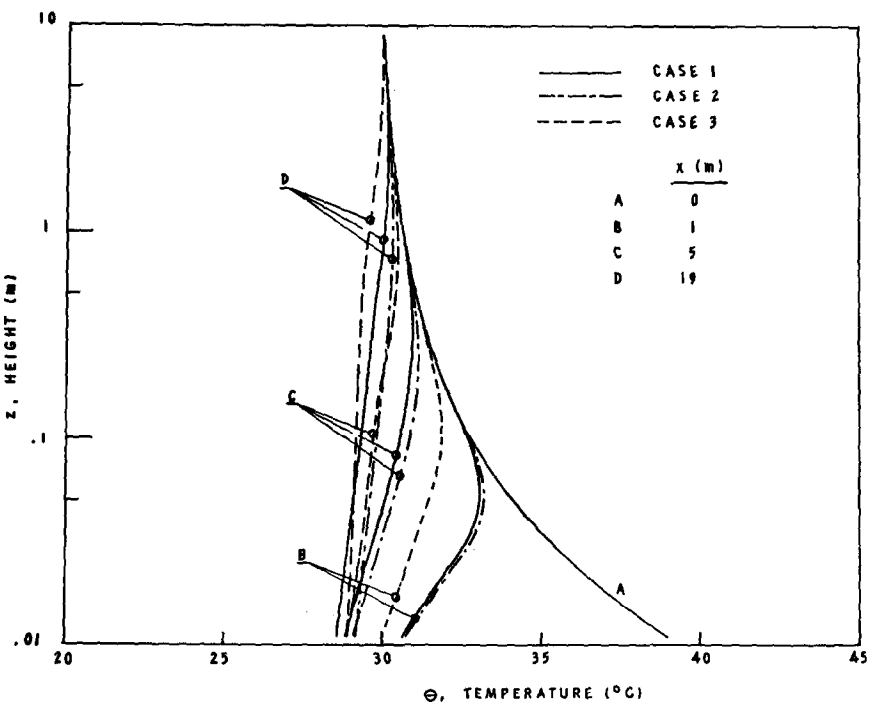


Fig. 7. Effect of variation of surface relative humidity on surface evaporation rate and the Bowen ratio.



ig. 8. Effect of thermal stratification and roughness change on calculated mean temperature profiles. (See text for further explanation.)

and the observations (see Rider *et al.*, 1965). It is interesting and useful to analyze the sensitivity of the predicted profiles to the roughness change and thermal stratification effects (see Figures 8 to 11), as follows: *Case 1* indicates the predictions of the present model, using the flow parameters given before, with the roughness change and thermal stratification taken into account. *Case 2* includes the roughness change, but thermal stratification is neglected by setting the buoyancy terms in the turbulent flux Equations (2), (3) and (4) identically equal to zero. This implies that temperature and moisture behave as passive scalars in the flow. *Case 3* includes thermal stratification, but the

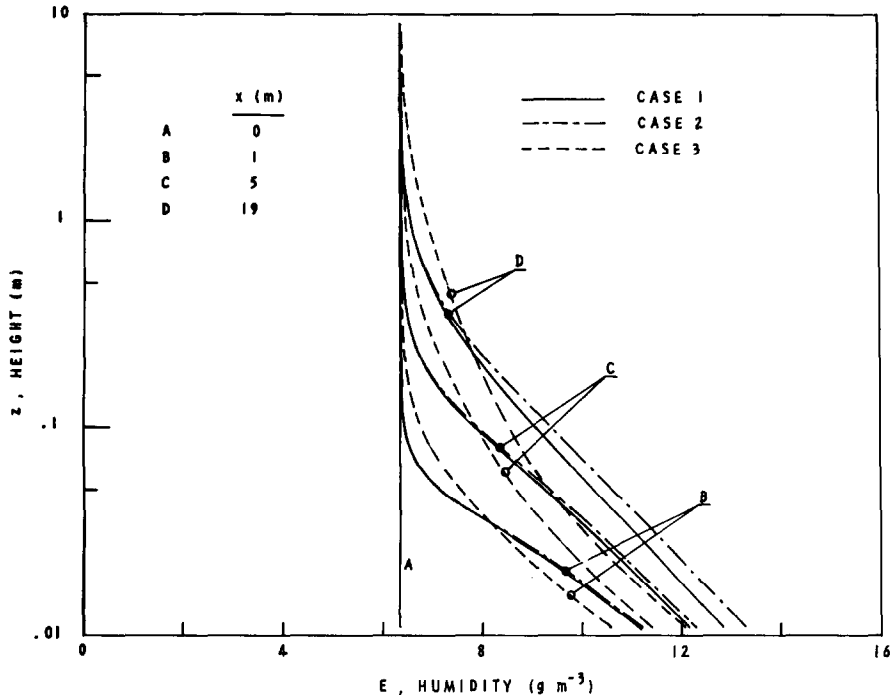


Fig. 9. Effect of thermal stratification and roughness change on calculated mean humidity profiles. (See text for further explanation.)

roughness change is eliminated by setting $z_{01} = z_{02}$. All other flow conditions were kept identical in the three cases.

Figures 8 and 9 show the effects of roughness change and thermal stratification on the calculated mean temperature and mean humidity profiles, respectively. It can be seen that the neglect of stratification makes only a small difference since the present study deals with the atmosphere closest to the ground where the buoyancy effects are expected to be small. As the thickness of the internal boundary layer increases with fetch, thermal stratification becomes important at large x. On the other hand, neglect of the roughness change has the largest influence on predictions closest to the surface

transition. As the fetch increases, the effect of the roughness change becomes less important. The two effects influence the predictions in opposite ways.

The calculated turbulent heat flux ($\overline{\theta w}$) and humidity flux ($\overline{ew}$) profiles are shown in Figures 10 and 11, respectively. The heat flux distributions clearly show the growth of the inversion layer with fetch. The thickness of this layer is given by the height where the heat flux goes through zero as it changes sign. The upstream turbulent fluxes are invariant with height. A new equilibrium layer develops downstream in which the fluxes are nearly constant, and the thickness of this layer increases with fetch. The effects of neglecting buoyancy (case 2) or neglecting the roughness change (case 3) on

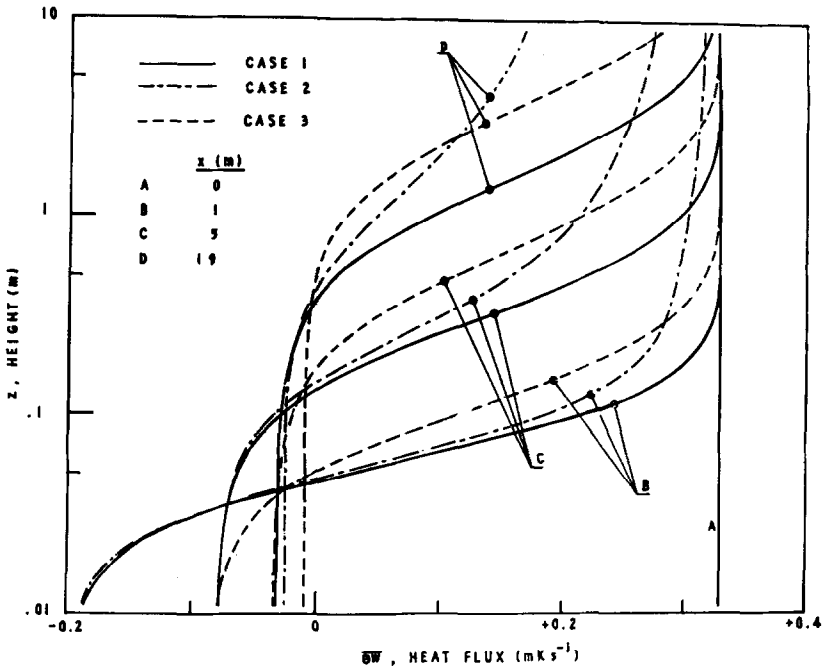


Fig. 10. Effect of thermal stratification and roughness change on calculated heat flux profiles. (See text for further explanation.)

the predicted turbulent flux profiles are also shown in the figures. The effect of roughness change on the mean wind and the turbulent shear stress profiles has been discussed by Rao *et al.* (1974).

The available energy in Equation (16) can be expressed (de Vries, 1959) as

$$R_N - G = (1 - r)\, R_s + R_a - \xi\sigma\Theta_0^4 - G, \tag{21}$$

where R_s and R_a are the intensities of short-wave and atmospheric radiation at the surface, r is the reflection coefficient of surface for short-wave radiation, ξ is the surface emissivity, and σ is the Stefan-Boltzmann Constant. It is generally assumed that horizontal changes in R_s and R_a can be neglected when the irrigated area is of

limited extent. Since the irrigated surface is at a lower temperature than the dry surface, $\xi\sigma\Theta_0^4$ is smaller, and the net radiation (R_N) is larger downstream. The data of Rider *et al.* (1963) show that the increase in net radiation for the wet surface is of the order of 2.5 mcal cm^{-2} s^{-1}. On the other hand, the smaller temperature gradient in the upper layers of the soil gives a smaller soil heat flux (G) in the wet area, despite the partial compensation provided by the greater thermal conductivity of the wet soil. Hence we expect the available energy ($R_N - G$) downstream to be larger than that upstream (H_{01}).

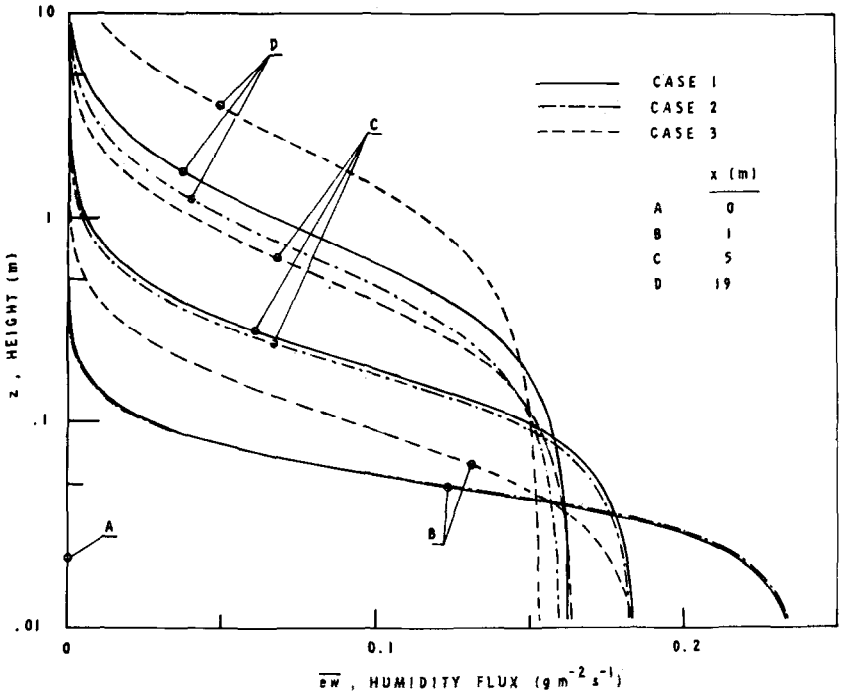


Fig. 11. Effect of thermal stratification and roughness change on calculated humidity flux profiles. (See text for further explanation.)

Figure 12 shows the calculated mean temperature and humidity profiles of *Case 4* for which $R_N - G = 1.2\,H_{01}$. The experimental data, and the profiles of case 1 for which $R_N - G = H_{01}$, are also shown in the figure for comparison. As 20% more energy becomes available for partition, the calculated temperature and humidity of the air increase, and the fit to the data improves.

5. Conclusions

The good agreement between the measured and predicted temperature and humidity

profiles is encouraging. The same model has given equally good agreement in the neutral step-change-of-roughness problem (Rao *et al.*, 1974). Unfortunately in both cases there are no detailed turbulence measurements within the internal boundary layer, which would provide a more demanding test of the predictions. However, similar turbulence models have been tested extensively in laboratory flows (see, e.g., Hanjalic and Launder, 1972), and the turbulence predictions there agree quite well with measurements.

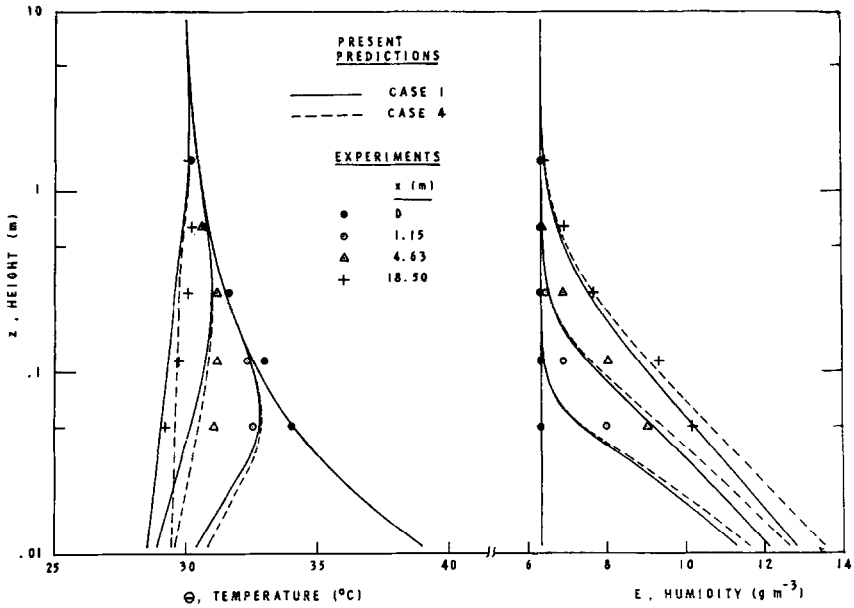


Fig. 12. Effect of a 20% increase in available energy ($R_N - G$) for wet surface on calculated mean temperature and humidity profiles. (See text for further explanation.)

We emphasize that our model uses relatively simple closure expressions that are in the nature of first approximations. All models of this type are in an evolutionary stage, and closure refinements are to be expected as new flows are modeled. Our results should therefore not be taken as definitive predictions, but rather as examples of the output of a current model.

Measurements of surface-layer structure under inhomogeneous conditions are rare. Most research to date has been concerned with the homogeneous case, which is now so well documented (Haugen, 1973) that, we can hope, experimenters will expand some of the pioneering work (Rider *et al.*, 1963; Morgan *et al.*, 1970) to inhomogeneous flows. Turbulence model studies can help not only in interpreting the results of such experiments, but in designing them as well. Preliminary designs can be run on the computer to assess the sensitivity to various parameters, for example, and to decide

what should be measured in the field. The experimental results, of course, would be invaluable in testing and refining the model.

Acknowledgements

We are grateful to our colleagues in the Boundary Layer Branch of AFCRL for helpful suggestions in the course of this work. We wish to thank Mr Anthony J. Matthews for assisting in the preparation of the figures. One of us (K.S.R.) would like to express his sincere appreciation to the National Research Council and the National Academy of Sciences for their support.

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